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Probability and Stochastic Processes

CAT 3

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Q1. A clinic has one doctor, and patients arrive according to a Poisson process at a rate of 6 patients/hour. The service time follows an exponential distribution with an average service rate of 8 patients/hour.

Patients arrive to a single-server clinic according to a Poisson process with rate $\lambda = 6$ patients/hour.

Service times are i.i.d. exponential with rate $\mu = 8$ patients/hour.

Stability (ergodicity) condition. For an $M/M/1$ queue to reach steady state we require $\lambda < \mu$. Here, $6 < 8$, so the system is stable.

Define the traffic intensity

$$\rho = \frac{\lambda}{\mu} = \frac{6}{8} = 0.75.$$

For the steady state.

Steady-state distribution of number in system. For $M/M/1$,

$$\mathbb{P}\{N = n\} = (1 - \rho) \rho^n, \quad n = 0, 1, 2, \dots$$

and the empty-system probability is $P_0 = \mathbb{P}\{N = 0\} = 1 - \rho$.

Performance Measures

Throughout, we will use the standard $M/M/1$ formulas and also cross-check with Little's Law:

$$L = \lambda W \quad \text{and} \quad L_q = \lambda W_q.$$

(a) Utilization

$$\rho = \frac{\lambda}{\mu} = \frac{6}{8} = 0.75 = 75\%.$$

(b) Mean number in system L

For $M/M/1$,

$$L = \frac{\rho}{1 - \rho}.$$

Substitute $\rho = 0.75$:

$$L = \frac{0.75}{1 - 0.75} = \frac{0.75}{0.25} = 3.00 \text{ patients.}$$

(c) Mean number in queue L_q

For $M/M/1$,

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{\lambda^2}{\mu(\mu - \lambda)}.$$

Compute via ρ :

$$\rho^2 = (0.75)^2 = 0.5625, \quad L_q = \frac{0.5625}{0.25} = 2.25 \text{ patients.}$$

(Cross-check using rates: $L_q = \frac{6^2}{8(8-6)} = \frac{36}{16} = 2.25$.)

(d) Mean waiting time in system W

For $M/M/1$,

$$W = \frac{1}{\mu - \lambda}.$$

Compute:

$$\mu - \lambda = 8 - 6 = 2 \text{ hr}^{-1}, \quad W = \frac{1}{2} = 0.5 \text{ hr} = 30 \text{ minutes}.$$

Little's Law cross-check: $L = \lambda W \Rightarrow 3 = 6 \times 0.5$, correct.

(e) Mean waiting time in queue W_q

For $M/M/1$,

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{\lambda}{\mu(\mu - \lambda)}.$$

Compute with ρ :

$$W_q = \frac{0.75}{2} = 0.375 \text{ hr}.$$

Convert to minutes:

$$0.375 \times 60 = 22.5 \text{ minutes}.$$

Little's Law cross-check: $L_q = \lambda W_q \Rightarrow 2.25 = 6 \times 0.375$, correct.

(f) Probability system is empty P_0

$$P_0 = 1 - \rho = 1 - 0.75 = 0.25 = 25\%.$$

(g) Probability of more than 4 customers in system

Using the geometric tail for $M/M/1$:

$$\mathbb{P}\{N > n\} = \rho^{n+1}, \quad n \geq 0.$$

Set $n = 4$:

$$\mathbb{P}\{N > 4\} = \rho^5 = (0.75)^5.$$

Compute stepwise:

$$0.75^2 = 0.5625, \quad 0.75^3 = 0.5625 \times 0.75 = 0.421875, \quad 0.75^4 = 0.421875 \times 0.75 = 0.31640625, \quad 0.75^5 = 0.31640625 \times 0.75 =$$

Therefore,

$$\mathbb{P}\{N > 4\} \approx 0.2373046875 \approx 23.73\%.$$

(Equivalently, $\mathbb{P}\{N \leq 4\} = 1 - 0.75^5 \approx 0.7626953125$, i.e., 76.27%.)

Q2. At a help desk, customers arrive at a rate of 3 per min, and the clerk can serve them at an average rate of 12 per hour.

Customers arrive according to a Poisson process with rate

$$\lambda = 3 \text{ customers/min.}$$

Service times are i.i.d. exponential with average service *rate*

$$\mu = 12 \text{ customers/hour} = \frac{12}{60} \text{ customers/min} = 0.2 \text{ customers/min.}$$

This is an $M/M/1$ queue. We must first check the **stability condition** for steady state to exist:

$$\lambda < \mu \quad (\text{required for stationarity}).$$

Here,

$$\lambda = 3 \text{ min}^{-1}, \quad \mu = 0.2 \text{ min}^{-1},$$

so

$$\lambda > \mu \quad \Rightarrow \quad \text{the system is } \textit{unstable}.$$

Traffic intensity (utilization).

$$\rho = \frac{\lambda}{\mu} = \frac{3}{0.2} = 15.$$

Since $\rho > 1$, the queue is unstable: the expected number in system grows without bound over time and the usual steady-state $M/M/1$ formulas do *not* apply (they require $\rho < 1$).

Implications for Requested Quantities

For an $M/M/1$ queue with $\rho \geq 1$, steady-state measures are undefined (infinite). Concretely:

- (a) Utilization $\rho = 15$ (i.e., 1500%, indicates overload).
- (b) Mean number in system L does not exist (diverges to ∞).
- (c) Mean number in queue L_q does not exist (diverges to ∞).
- (d) Mean waiting time in system W does not exist (diverges to ∞).
- (e) Mean waiting time in queue W_q does not exist (diverges to ∞).
- (f) Probability system empty $P_0 = 0$ in the long run (server essentially never idle).
- (g) Probability of more than 4 customers in system $\mathbb{P}\{N > 4\} = 1$ in the long run.

- **Unit conversion is crucial.** The apparent overload arises because arrivals are very fast (3/min = 180/hour) while service is very slow (12/hour).
- **Design insight.** To achieve stability, one must have $\mu > \lambda$. In this case, either increase service rate beyond 3 per minute ($\mu > 3/\text{min}$) by adding servers or speeding service, or reduce arrival rate (e.g., appointment system, triage).
- If, hypothetically, the intended service rate were 12 *per minute* (not per hour), then $\mu = 12/\text{min}$ and $\rho = 3/12 = 0.25 < 1$, and the standard finite $M/M/1$ formulas would apply.

$\rho = 15, \quad L = \infty, \quad L_q = \infty, \quad W = \infty, \quad W_q = \infty, \quad P_0 = 0, \quad \mathbb{P}\{N > 4\} = 1.$
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Q3. Server utilization-A single-lane toll booth receives cars at a rate of 1 per hour, and the operator can process cars at 20 per min. Compute the

- Arrival rate: $\lambda = 1$ car/hour.
- Service rate given: 20 cars/min.
- Convert service rate to per hour: $\mu = 20 \times 60 = 1200$ cars/hour.

This is an $M/M/1$ queue (Poisson arrivals, exponential service, single server, FCFS, infinite buffer).

Stability check. For steady state, need $\lambda < \mu$. Here $1 < 1200$, so the system is stable.

Traffic intensity (utilization).

$$\rho = \frac{\lambda}{\mu} = \frac{1}{1200} = 0.000833\bar{3} \approx 0.08333\%.$$

Steady-state distribution (for reference). For $M/M/1$,

$$\mathbb{P}\{N = n\} = (1 - \rho)\rho^n, \quad n = 0, 1, 2, \dots$$

and tail probability $\mathbb{P}\{N > n\} = \rho^{n+1}$.

Performance Measures

We will use the standard $M/M/1$ formulas and cross-check with Little's Law ($L = \lambda W$, $L_q = \lambda W_q$).

(a) Utilization

$$\rho = \frac{1}{1200} \approx 0.000833\bar{3} \quad (\approx 0.08333\%).$$

(b) Mean number in system L

$$L = \frac{\rho}{1 - \rho} = \frac{\frac{1}{1200}}{1 - \frac{1}{1200}} = \frac{1/1200}{1199/1200} = \frac{1}{1199} \approx 0.0008340283569641369.$$

$$L = \frac{1}{1199} \approx 8.34028 \times 10^{-4} \text{ cars.}$$

(c) Mean number in queue L_q

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{\left(\frac{1}{1200}\right)^2}{1 - \frac{1}{1200}} = \frac{\frac{1}{1,440,000}}{\frac{1199}{1200}} = \frac{1}{1,440,000} \cdot \frac{1200}{1199} = \frac{1}{1200 \cdot 1199} = \frac{1}{1,438,800}.$$

Numerically,

$$L_q = \frac{1}{1,438,800} \approx 6.950236308 \times 10^{-7} \text{ cars.}$$

(d) Mean waiting time in system W

$$W = \frac{1}{\mu - \lambda} = \frac{1}{1200 - 1} = \frac{1}{1199} \text{ hours.}$$

Convert units:

$$W = \frac{1}{1199} \text{ hr} \approx 0.0008340284 \text{ hr} \approx 0.0500417 \text{ min} \approx \frac{3600}{1199} \text{ s} \approx 3.002502085 \text{ s.}$$

$$W = \frac{1}{1199} \text{ hr} \approx 3.0025 \text{ s.}$$

(*Cross-check:* $L = \lambda W \Rightarrow \frac{1}{1199} = 1 \cdot \frac{1}{1199}$, OK.)

(e) Mean waiting time in queue W_q

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{\frac{1}{1200}}{1199} = \frac{1}{1200 \cdot 1199} = \frac{1}{1,438,800} \text{ hours.}$$

Convert units:

$$W_q \approx 6.950236308 \times 10^{-7} \text{ hr} \approx 2.502085086 \times 10^{-5} \text{ min} \approx 0.002502085086 \text{ s (about 2.50 ms).}$$

$$W_q = \frac{1}{1,438,800} \text{ hr} \approx 0.00250 \text{ s.}$$

(*Cross-check:* $L_q = \lambda W_q \Rightarrow \frac{1}{1,438,800} = 1 \cdot \frac{1}{1,438,800}$, OK.)

(f) Probability system empty P_0

$$P_0 = 1 - \rho = 1 - \frac{1}{1200} = \frac{1199}{1200} \approx 0.999166\bar{6}.$$

(g) Probability of more than 4 customers in system

$$\mathbb{P}\{N > 4\} = \rho^5 = \left(\frac{1}{1200}\right)^5 \approx 4.018775720 \times 10^{-16}.$$

$$\mathbb{P}\{N > 4\} \approx 4.02 \times 10^{-16} \text{ (essentially zero).}$$

Q4. In a small post office, arrivals occur at a rate of 5 per hour, and service rate is 6 per hour. Compute the

Arrivals follow a Poisson process with rate $\lambda = 5$ per hour. Service times are i.i.d. exponential with rate $\mu = 6$ per hour. This is a single-server $M/M/1$ queue.

Stability (ergodicity) check. For an $M/M/1$ to reach steady state we require $\lambda < \mu$. Here $5 < 6$, so the system is stable.

Traffic intensity (utilization).

$$\rho = \frac{\lambda}{\mu} = \frac{5}{6} = 0.8333\bar{3} \quad (\approx 83.33\%).$$

Steady-state distribution (reference). For $M/M/1$,

$$\mathbb{P}\{N = n\} = (1 - \rho)\rho^n, \quad n = 0, 1, 2, \dots, \quad \mathbb{P}\{N > n\} = \rho^{n+1}.$$

Performance Measures

We use standard $M/M/1$ formulas and cross-check with Little's Law ($L = \lambda W$ and $L_q = \lambda W_q$).

(a) Utilization

$$\rho = \frac{5}{6} \approx 0.8333\bar{3} \text{ (83.33\%).}$$

(b) Mean number in system L

$$L = \frac{\rho}{1 - \rho} = \frac{\frac{5}{6}}{1 - \frac{5}{6}} = \frac{\frac{5}{6}}{\frac{1}{6}} = 5.$$
$$L = 5 \text{ customers.}$$

(c) Mean number in queue L_q

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{\left(\frac{5}{6}\right)^2}{\frac{1}{6}} = \frac{\frac{25}{36}}{\frac{1}{6}} = \frac{25}{36} \cdot 6 = \frac{25}{6} \approx 4.1667.$$
$$L_q = \frac{25}{6} \approx 4.1667 \text{ customers.}$$

(d) Mean waiting time in system W

$$W = \frac{1}{\mu - \lambda} = \frac{1}{6 - 5} = 1 \text{ hour} = 60 \text{ minutes.}$$

(Cross-check: $L = \lambda W \Rightarrow 5 = 5 \times 1$, OK.)

$$W = 1 \text{ hr} = 60 \text{ min.}$$

(e) Mean waiting time in queue W_q

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{\frac{5}{6}}{1} = \frac{5}{6} \text{ hour} \approx 0.8333 \text{ hr} = 50 \text{ minutes.}$$

(Cross-check: $L_q = \lambda W_q \Rightarrow \frac{25}{6} = 5 \times \frac{5}{6}$, OK.)

$$W_q = \frac{5}{6} \text{ hr} \approx 50 \text{ min.}$$

(f) Probability system empty P_0

$$P_0 = 1 - \rho = 1 - \frac{5}{6} = \frac{1}{6} \approx 0.1667 \quad (16.67\%).$$

$$P_0 = \frac{1}{6} \approx 0.1667.$$

(g) Probability of more than 4 customers in the system

Use the geometric tail:

$$\mathbb{P}\{N > 4\} = \rho^5 = \left(\frac{5}{6}\right)^5 = \frac{3125}{7776} \approx 0.4018775720.$$

Thus about 40.19%.

$$\mathbb{P}\{N > 4\} = \left(\frac{5}{6}\right)^5 \approx 0.40188 \quad (40.19\%).$$

Q5. In a call center with one operator, calls arrive at a rate of 2 per minute, and the operator serves at 3 per minute. Compute the

Calls arrive according to a Poisson process with rate

$$\lambda = 2 \text{ per minute,}$$

and service times are i.i.d. exponential with rate

$$\mu = 3 \text{ per minute.}$$

This is a single-server $M/M/1$ queue (FCFS, infinite buffer).

Stability check. Steady state exists if $\lambda < \mu$. Here $2 < 3$, so the system is stable.

Traffic intensity (utilization).

$$\rho = \frac{\lambda}{\mu} = \frac{2}{3} = 0.666\bar{6} \quad (\text{about } 66.67\%).$$

Steady-state distribution (reference). For $M/M/1$,

$$\mathbb{P}\{N = n\} = (1 - \rho)\rho^n, \quad n = 0, 1, 2, \dots \quad \text{and} \quad \mathbb{P}\{N > n\} = \rho^{n+1}.$$

Performance Measures

We use the standard $M/M/1$ formulas and cross-check with Little's Law ($L = \lambda W$, $L_q = \lambda W_q$).

(a) Utilization

$$\rho = \frac{2}{3} \approx 0.6667 \text{ (66.67\%).}$$

(b) Mean number in system L

$$L = \frac{\rho}{1 - \rho} = \frac{\frac{2}{3}}{1 - \frac{2}{3}} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2.$$

$$L = 2 \text{ calls.}$$

(c) Mean number in queue L_q

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{\left(\frac{2}{3}\right)^2}{\frac{1}{3}} = \frac{\frac{4}{9}}{\frac{1}{3}} = \frac{4}{9} \cdot 3 = \frac{4}{3} \approx 1.3333.$$

$$L_q = \frac{4}{3} \approx 1.3333 \text{ calls.}$$

(d) Mean waiting time in system W

$$W = \frac{1}{\mu - \lambda} = \frac{1}{3 - 2} = 1 \text{ minute} = 60 \text{ seconds.}$$

Cross-check (Little's Law): $L = \lambda W \Rightarrow 2 = 2 \times 1$, correct.

$$W = 1 \text{ min} = 60 \text{ s.}$$

(e) Mean waiting time in queue W_q

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{\frac{2}{3}}{1} = \frac{2}{3} \text{ minute.}$$

Convert to seconds:

$$\frac{2}{3} \text{ min} \times 60 \text{ s/min} = 40 \text{ s.}$$

Cross-check (Little's Law): $L_q = \lambda W_q \Rightarrow \frac{4}{3} = 2 \times \frac{2}{3}$, correct.

$$W_q = \frac{2}{3} \text{ min} = 40 \text{ s.}$$

(f) Probability system is empty P_0

$$P_0 = 1 - \rho = 1 - \frac{2}{3} = \frac{1}{3} \approx 0.3333 \quad (33.33\%).$$

$$P_0 = \frac{1}{3} \approx 0.3333.$$

(g) Probability of more than 4 customers in the system

Use the geometric tail:

$$\mathbb{P}\{N > 4\} = \rho^5 = \left(\frac{2}{3}\right)^5 = \frac{32}{243}.$$

Numerically:

$$\frac{32}{243} \approx 0.1316872428 \approx 13.17\%.$$

$$\mathbb{P}\{N > 4\} = \left(\frac{2}{3}\right)^5 = \frac{32}{243} \approx 0.13169 \quad (13.17\%).$$

Q6. Using Chapman-Kolmogorov assumptions, derive Poisson process, hence compute the mean and the variance the resulting Poisson Proce (Show all your working)

Let $\{N(t) : t \geq 0\}$ be a counting process with the following (standard) assumptions:

1. **Initial condition:** $N(0) = 0$ almost surely.
2. **Independent increments:** For $0 \leq t_0 < t_1 < \dots < t_k$, the increments $N(t_1) - N(t_0), \dots, N(t_k) - N(t_{k-1})$ are independent.
3. **Stationary increments:** The law of $N(t + s) - N(s)$ depends only on t (not on s).
4. **Orderliness (small-interval behavior):** There exists $\lambda > 0$ such that, as $h \downarrow 0$,

$$\mathbb{P}\{N(h) = 1\} = \lambda h + o(h), \quad (1)$$

$$\mathbb{P}\{N(h) \geq 2\} = o(h), \quad (2)$$

$$\mathbb{P}\{N(h) = 0\} = 1 - \lambda h + o(h). \quad (3)$$

Assumptions 2–3 are exactly those that justify a Chapman–Kolmogorov *convolution* relation for the distributions of the counts, and Assumption 4 (orderliness) ensures at most one arrival occurs with probability $\Theta(h)$ over small intervals.

Denote $p_n(t) := \mathbb{P}\{N(t) = n\}$.

Chapman–Kolmogorov (Convolution) Equation

By *independent* and *stationary* increments,

$$\mathbb{P}\{N(t + s) = n\} = \sum_{k=0}^n \mathbb{P}\{N(s) = k\} \mathbb{P}\{N(t + s) - N(s) = n - k\} = \sum_{k=0}^n p_k(s) p_{n-k}(t).$$

Thus for all $s, t \geq 0$ and $n \geq 0$,

$$p_n(t + s) = \sum_{k=0}^n p_k(s) p_{n-k}(t). \quad (4)$$

Deriving the Kolmogorov Forward (Differential–Difference) Equations

Fix $t \geq 0$ and consider a small increment $h > 0$. Using the law of total probability (conditioning on $N(t)$) together with the small-interval probabilities (1)–(3):

Case $n = 0$:

$$p_0(t + h) = p_0(t) \mathbb{P}\{N(t + h) - N(t) = 0\} + \underbrace{\sum_{k \geq 1} p_k(t) \mathbb{P}\{N(h) = -k\}}_{=0} = p_0(t) (1 - \lambda h + o(h)).$$

Hence

$$\frac{p_0(t + h) - p_0(t)}{h} = -\lambda p_0(t) + o(1) \implies p'_0(t) = -\lambda p_0(t), \quad p_0(0) = 1.$$

Case $n \geq 1$: In $[t, t+h)$, either (i) no arrival occurs and we stay at n , or (ii) exactly one arrival occurs and we move from $n-1$ to n . Multiple arrivals are $o(h)$:

$$p_n(t+h) = p_n(t)(1 - \lambda h + o(h)) + p_{n-1}(t)(\lambda h + o(h)) + o(h).$$

Therefore

$$\frac{p_n(t+h) - p_n(t)}{h} = -\lambda p_n(t) + \lambda p_{n-1}(t) + o(1) \implies p'_n(t) = -\lambda p_n(t) + \lambda p_{n-1}(t), \quad n \geq 1,$$

with $p_n(0) = 0$ for $n \geq 1$.

We have obtained the **Kolmogorov forward equations** (for a pure birth process with birth rate λ):

$$p'_0(t) = -\lambda p_0(t), \quad p_0(0) = 1, \quad (5)$$

$$p'_n(t) = -\lambda p_n(t) + \lambda p_{n-1}(t), \quad p_n(0) = 0, \quad n \geq 1. \quad (6)$$

Solving the System (Two Routes)

Route A: Probability Generating Function (PGF)

Define $G(z, t) := \sum_{n=0}^{\infty} p_n(t) z^n$ for $|z| \leq 1$. Multiply (6) by z^n and sum over $n \geq 1$, and use (5) for $n = 0$:

$$\frac{\partial}{\partial t} G(z, t) = \sum_{n=0}^{\infty} p'_n(t) z^n = -\lambda \sum_{n=0}^{\infty} p_n(t) z^n + \lambda \sum_{n=1}^{\infty} p_{n-1}(t) z^n = -\lambda G(z, t) + \lambda z G(z, t).$$

Hence

$$\frac{\partial}{\partial t} G(z, t) = \lambda(z-1) G(z, t), \quad G(z, 0) = 1.$$

Solve the linear ODE in t :

$$G(z, t) = \exp(\lambda t(z-1)).$$

Expanding in powers of z :

$$G(z, t) = e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} z^n \implies p_n(t) = \mathbb{P}\{N(t) = n\} = \boxed{\frac{e^{-\lambda t} (\lambda t)^n}{n!}}, \quad n = 0, 1, 2, \dots$$

Thus $N(t) \sim \text{Poisson}(\lambda t)$ for each fixed t .

Route B: Induction from the Forward Equations

From (5), $p_0(t) = e^{-\lambda t}$. For $n \geq 1$, write (6) as

$$\frac{d}{dt} (e^{\lambda t} p_n(t)) = \lambda e^{\lambda t} p_{n-1}(t).$$

Integrate from 0 to t and use $p_n(0) = 0$:

$$e^{\lambda t} p_n(t) = \lambda \int_0^t e^{\lambda s} p_{n-1}(s) ds.$$

Assuming $p_{n-1}(s) = e^{-\lambda s} (\lambda s)^{n-1} / (n-1)!$, we get

$$e^{\lambda t} p_n(t) = \lambda \int_0^t \frac{(\lambda s)^{n-1}}{(n-1)!} ds = \lambda \cdot \frac{\lambda^{n-1}}{(n-1)!} \cdot \frac{t^n}{n} = \frac{\lambda^n t^n}{n!}.$$

Hence $p_n(t) = e^{-\lambda t} (\lambda t)^n / n!$ for all n .

Mean and Variance of the Poisson Process

Since $N(t) \sim \text{Poisson}(\lambda t)$, its PGF is $G(z, t) = \exp(\lambda t(z - 1))$ and its MGF is

$$M_{N(t)}(\theta) = \mathbb{E}[e^{\theta N(t)}] = G(e^\theta, t) = \exp(\lambda t(e^\theta - 1)).$$

Differentiate $\log M$ to obtain cumulants, or differentiate M directly:

Mean.

$$\mathbb{E}[N(t)] = M'_{N(t)}(0) = \exp(\lambda t(e^0 - 1)) \cdot \lambda t e^0 = \boxed{\lambda t}.$$

Variance.

$$\text{Var}(N(t)) = M''_{N(t)}(0) - (M'_{N(t)}(0))^2 = (\lambda t e^0 + (\lambda t e^0)^2) - (\lambda t)^2 = \boxed{\lambda t}.$$

Alternative (moment ODE cross-check). From the forward equations one can show

$$\frac{d}{dt} \mathbb{E}[N(t)] = \lambda, \quad \mathbb{E}[N(0)] = 0 \Rightarrow \mathbb{E}[N(t)] = \lambda t,$$

and

$$\frac{d}{dt} \mathbb{E}[N(t)^2] = 2\lambda \mathbb{E}[N(t)] + \lambda = 2\lambda(\lambda t) + \lambda,$$

hence $\mathbb{E}[N(t)^2] = \lambda^2 t^2 + \lambda t$, giving $\text{Var}(N(t)) = \mathbb{E}[N^2] - \mathbb{E}[N]^2 = \lambda t$.

Conclusion

Under the Chapman–Kolmogorov assumptions (independent & stationary increments) with orderly small-interval behavior (1)–(2), the counting process $\{N(t)\}$ is characterized by the forward equations (5)–(6) whose solution is

$$\boxed{N(t) \sim \text{Poisson}(\lambda t)}.$$

Consequently,

$$\boxed{\mathbb{E}[N(t)] = \lambda t, \quad \text{Var}(N(t)) = \lambda t.}$$